

Since the number of mismatches is too low, to identify a mismatch and to study the performance it is better to use the `table()` function over the difference between the test and predicted values.

```
>performance <- test.df[,1] == predict.dl2[,1]

>table(performance)
```

```
performance
FALSE TRUE
222 3778
```

Performance shows that, out of 4000 test data cases, deep neural net failed to identify the correct letter only in 222 cases.

To compare the performance with other methods it is better to have a percentage evaluation of the classification task. Here is a simple way to do this.

```
>table(performance)*100/4000

FALSE TRUE
5.55 94.45
```

The result shows that the success rate of this experiment is 94.45 per cent.

In classification exercises, it is always better to have a comparative study of methods. I have done the above classification task with Support Vector Machine to explore a better alternative and to study how good this method is for this task.

Using Support Vector Machine (SVM)

SVM is also useful to classify image data. I have used the SVM model to classify the same data to compare the performance between the deep learning and SVM models. Though the performance of individual models is highly dependent on different model parameters and there is every possibility to get different results in different runs, my objective here is to gauge the performance difference between these two supervised learning schemes. Readers may explore this further by applying this experiment to other neural network models. As in the earlier case, appropriate libraries are loaded and then the training and test data are prepared.

```
>library(kernlab)#load SVM package
>letters_train<- letterimage [1:16000, ]
>letters_test<- letterimage [16001:20000, ]
```

Next, train the SVM (`ksvm`) with the letter column as the only response variable and the rest of the columns as predictors. As Radial Basis Function (RBF) works better for classification tasks in many cases, I have used the RBF as the

SVM kernel function. Readers may use different functions, based on their preferences.

```
>letter_classifier_SVM<- ksvm(letter ~ ., data =
letters_train, kernel = "rbfdot")
```

To verify this SVM model, we can use the test data to predict the outcome of this data:

```
>letter_predictions<- predict(letter_classifier_
SVM, letters_test)
```

As in the earlier case, this performance can also be verified with the confusion matrix:

```
>agreement<- letter_predictions == letters_
test$letter
>table(agreement)
```

```
agreement
FALSE TRUE
643 3357
```

In percentage terms, this can be done in either of the following two ways.

1. By using the following command:

```
>table(agreement)*100/4000

agreement
FALSE TRUE
7.025 92.975
```

2. By using the `prop.table()` function.

From the classification results of H2O and SVM, it is apparent that both the supervised artificial neural network models methods are suitable for Optical Character Recognition classification and can be used to achieve higher performance. While H2O requires a more advanced computational platform, the requirements of SVM are less. But H2O provides a sophisticated versatile approach; so classifier models can be designed and monitored more flexibly than with the SVM approach. **END** 

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